

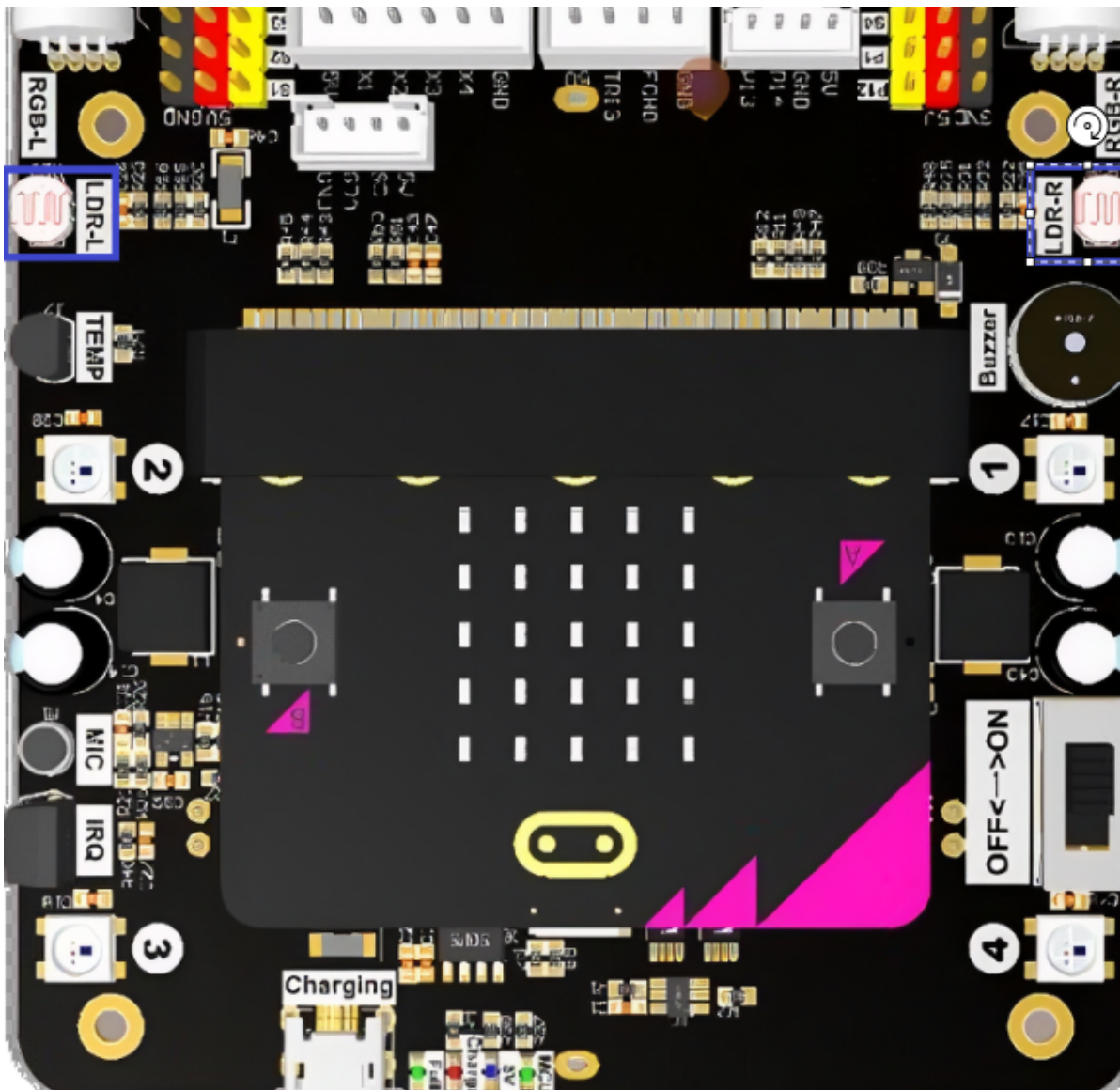
Moon Guardian

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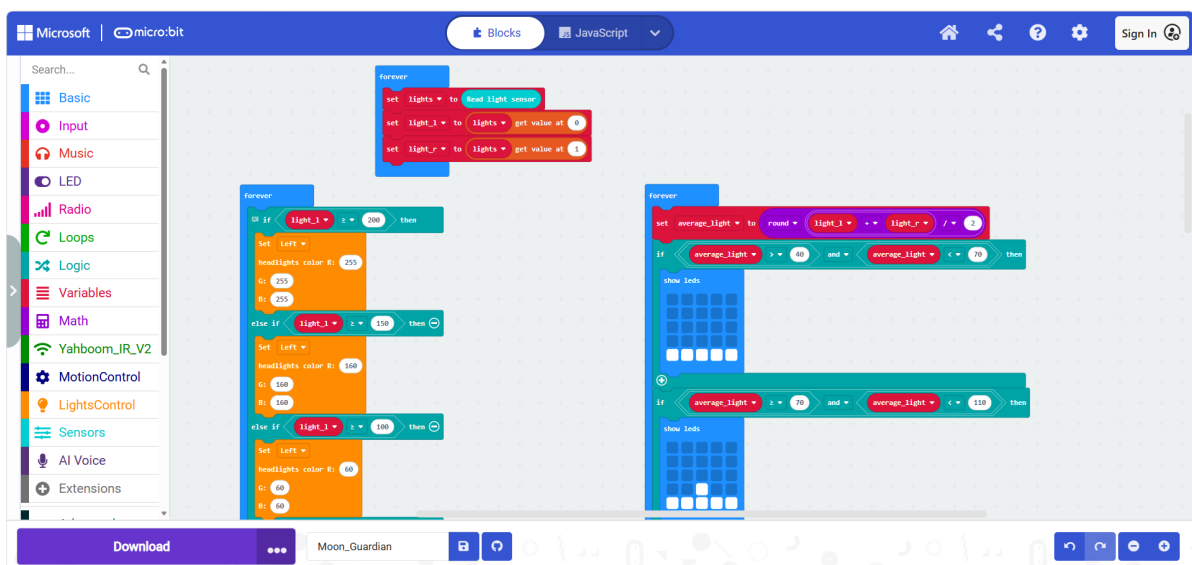
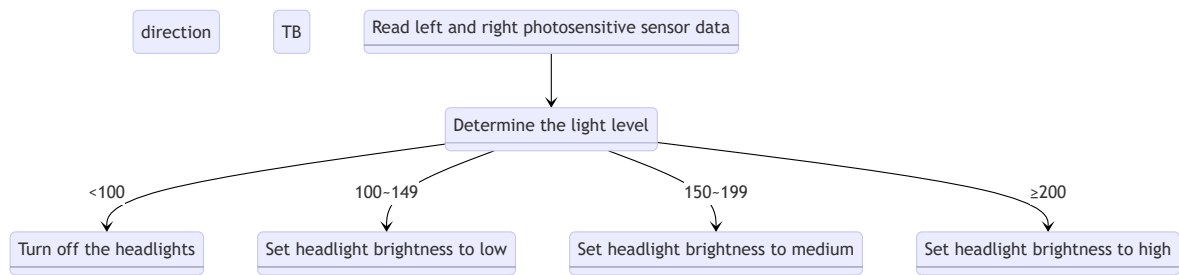
1. Experiment Purpose

In this section, we will use the photosensitive sensor together with light control to achieve the "moon guardian" effect.

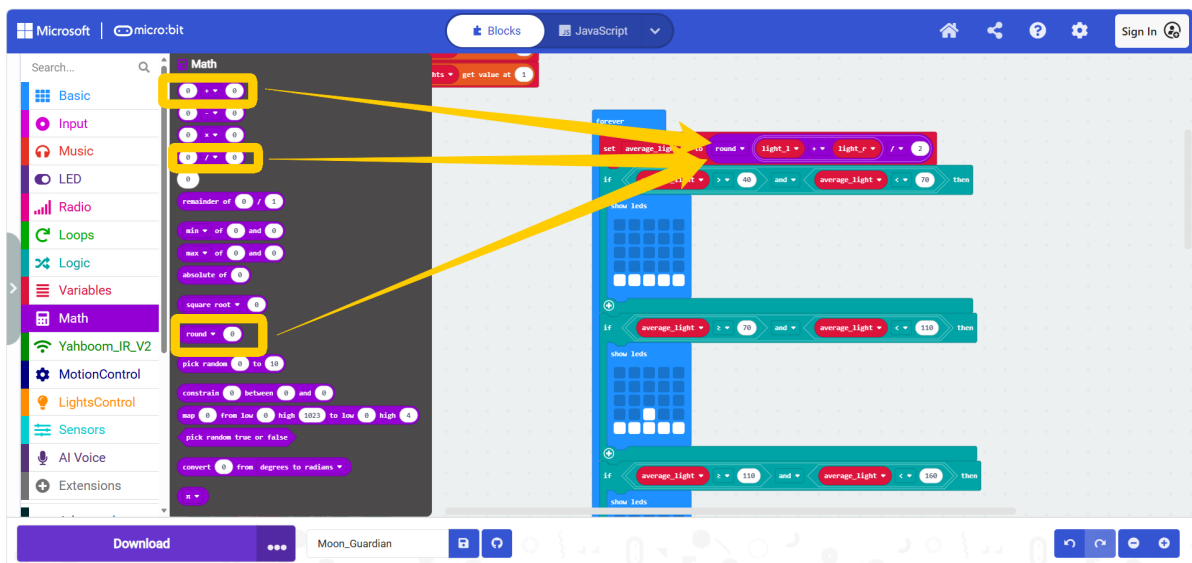


2. Implementation Principle

Thanks to the sensitivity of the photosensitive sensor to light, when the ambient light changes significantly, the read data also changes significantly. The darker the environment, the larger the read data. At this time, we can adjust the RGB parameters to adjust the brightness of the lights.



Addition, subtraction, multiplication, division, and rounding operations are located in the **Math** category.



The positions of the blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `19.Moon_Guardian.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

[!CAUTION]

Make sure your hands are dry; wet hands may cause a short circuit on the main board! Since the photosensitive sensor has errors, you can modify the photosensitive sensor thresholds according to the actual situation.

You can gradually cover the two photosensitive sensors with your hand, and you will find that the lights gradually brighten. You can also take the car to a darker environment to test. The darker the environment, the higher the pattern displayed on the Micro:bit LED matrix. Conversely, if you shine a flashlight on the photosensitive sensors, the car will turn off the lights, and the pattern shown on the Micro:bit LED matrix will be lower.

